Solve the quadratic equation $z^2 + (1-i)z - 3i = 0$.

Solution $a = 1,\ b = 1 - i$, and $c = -3i$ we have

$$z = \frac{-(1-i) + [(1-i)^2 - 4(-3i)]^{1/2}}{2} = \frac{1}{2}\left[-1 + i + (10i)^{1/2}\right]. \qquad ($$

To compute $(10i)^{1/2}$ $\theta = \pi/2$, and $n = 2,\ k = 0,\ k = 1$. The two square roots of $10i$ are:

r=10

$$w_0 = \sqrt{10}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right) = \sqrt{10}\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i\right) = \sqrt{5} + \sqrt{5}i$$

and

$$w_1 = \sqrt{10}\left(\cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4}\right) = \sqrt{10}\left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i\right) = -\sqrt{5} - \sqrt{5}i.$$

Therefore,

$$z_1 = \frac{1}{2}\left[-1 + i + \left(\sqrt{5} + \sqrt{5}i\right)\right] \quad \text{and} \quad z_2 = \frac{1}{2}\left[-1 + i + \left(-\sqrt{5} - \sqrt{5}i\right)\right].$$

These solutions of the original equation, written in the form $z = a + ib$, are

$$z_1 = \frac{1}{2}\left(\sqrt{5} - 1\right) + \frac{1}{2}\left(\sqrt{5} + 1\right)i \quad \text{and} \quad z_2 = -\frac{1}{2}\left(\sqrt{5} + 1\right) - \frac{1}{2}\left(\sqrt{5} - 1\right)i.$$

$$e^z = e^{x+iy} = e^x(\cos y + i\sin y).$$

$$e^{z_1}e^{z_2} = e^{z_1+z_2}.$$

$$z = re^{i\theta}.$$

This convenient form is called the **exponential form** of a complex number z. For example, $i = e^{\pi i/2}$ and $1 + i = \sqrt{2}e^{\pi i/4}$. Also, the formula for the n nth roots of a complex number,

$$z^{1/n} = \sqrt[n]{r}e^{i(\theta+2k\pi)/n}, k = 0, 1, 2, \ldots, n-1.$$

Complex Functions

*A **function** f from a set A to a set B is a rule of correspondence that assigns to each element in A one and only one element in B.*

We often think of a function as a rule or a machine that accepts *inputs* from the set A and returns *outputs* in the set B. In elementary calculus we studied functions whose inputs and outputs were real numbers. Such functions are called **real-valued functions of a real variable**. In this section we begin our study of functions whose inputs and outputs are complex numbers. Naturally, we call these functions **complex functions of a complex variable**, or **complex functions** for short.

Function Suppose that f is a function from the set A to the set B. If f assigns to the element a in A the element b in B, then we say that b is the **image** of a under f, or the **value** of f at a, and we write $b = f(a)$. The set A—the set of inputs—is called the **domain** of f and the set of images in B—the set of outputs—is called the **range** of f. We denote the domain and range of a function f by $\mathrm{Dom}(f)$ and $\mathrm{Range}(f)$, respectively. As an example, consider the "squaring" function $f(x) = x^2$ defined for the real variable x. Since any real number can be squared, the domain of f is the set $\mathbf{R}$ of all real numbers. That is, $\mathrm{Dom}(f) = A = \mathbf{R}$. The range of f consists of all real numbers x^2 where x is a real number. Of course, $x^2 \geq 0$ for all real x, and it is easy to see from the graph of f that the range of f is the set of all *nonnegative* real numbers. Thus, $\mathrm{Range}(f)$ is the interval $[0, \infty)$. The range of f need not be the same as the set B. For instance, because the interval $[0, \infty)$ is a subset of both $\mathbf{R}$ and the set $\mathbf{C}$ of all complex numbers, f can be viewed as a function from $A = \mathbf{R}$ to $B = \mathbf{R}$ or f can be viewed as a function from $A = \mathbf{R}$ to $B = \mathbf{C}$. In both cases, the range of f is contained in but not equal to the set B.

As the following definition indicates, a *complex* function is a function whose inputs and outputs are complex numbers.

Definition 2.1 Complex Function

A **complex function** is a function f whose domain and range are subsets of the set $\mathbf{C}$ of complex numbers.

A complex function is also called a **complex-valued function of a complex variable**. For the most part we will use the usual symbols f, g, and h to denote complex functions. In addition, inputs to a complex function f will typically be denoted by the variable z and outputs by the variable $w = f(z)$.

EXAMPLE 1 Complex Function

(a) The expression $z^2 - (2+i)z$ can be evaluated at any complex number z and always yields a single complex number, and so $f(z) = z^2 - (2+i)z$ defines a complex function. Values of f are found by using the arithmetic operations for complex numbers For instance, at the points $z = i$ and $z = 1+i$ we have:

$$f(i) = (i)^2 - (2+i)(i) = -1 - 2i + 1 = -2i$$

and $\quad f(1+i) = (1+i)^2 - (2+i)(1+i) = 2i - 1 - 3i = -1 - i.$

(b) The expression $g(z) = z + 2\,\mathrm{Re}(z)$ also defines a complex function. Some values of g are:

$$g(i) = i + 2\mathrm{Re}\,(i) = i + 2(0) = i$$

and $\quad g(2-3i) = 2 - 3i + 2\mathrm{Re}\,(2-3i) = 2 - 3i + 2(2) = 6 - 3i.$

When the domain of a complex function is not explicitly stated, we assume the domain to be the set of all complex numbers z for which $f(z)$ is defined. This set is sometimes referred to as the *natural domain* of f. For example

Real and Imaginary Parts of a Complex Function It is often helpful to express the inputs and the outputs a complex function in terms of their real and imaginary parts. If $w = f(z)$ is a complex function, then the image of a complex number $z = x + iy$ under f is a complex number $w = u + iv$. By simplifying the expression $f(x + iy)$, we can write the real

variables u and v in terms of the real variables x and y. For example, by replacing the symbol z with $x + iy$ in the complex function $w = z^2$, we obtain:

$$w = u + iv = (x + iy)^2 = x^2 - y^2 + 2xyi. \tag{1}$$

From (1) the real variables u and v are given by $u = x^2 - y^2$ and $v = 2xy$, respectively. This example illustrates that, if $w = u + iv = f(x + iy)$ is a complex function, then both u and v are real functions of the two real variables x and y. That is, by setting $z = x + iy$, we can express any complex function $w = f(z)$ in terms of two real functions as:

$$f(z) = u(x, y) + iv(x, y). \tag{2}$$

The functions $u(x,\ y)$ and $v(x,\ y)$ in (2) are called the **real and imaginary parts of f**, respectively.

Function of a Complex Variable

Let $w = f(z) = z^2 + 3z$. Find u and v and calculate the value of f at $z = 1 + 3i$.

Solution. $u = \operatorname{Re} f(z) = x^2 - y^2 + 3x$ and $v = 2xy + 3y$. Also,

$$f(1 + 3i) = (1 + 3i)^2 + 3(1 + 3i) = 1 - 9 + 6i + 3 + 9i = -5 + 15i.$$

This shows that $u(1, 3) = -5$ and $v(1, 3) = 15$. Check this by using the expressions for u and v.

Function of a Complex Variable

Let $w = f(z) = 2iz + 6\bar{z}$. Find u and v and the value of f at $z = \frac{1}{2} + 4i$.

Solution. $f(z) = 2i(x + iy) + 6(x - iy)$ gives $u(x, y) = 6x - 2y$ and $v(x, y) = 2x - 6y$. Also,

$$f(\tfrac{1}{2} + 4i) = 2i(\tfrac{1}{2} + 4i) + 6(\tfrac{1}{2} - 4i) = i - 8 + 3 - 24i = -5 - 23i.$$

$f(z)$ denotes the value of f at z,

.

We assume all functions to be ***single valued relations***, as usual: to each z in S there corresponds but *one* value $w = f(z)$ (but, of course, several z may give the same value $w = f(z)$, just as in calculus). Accordingly, we shall *not use* the term "multivalued function" (used in some books on complex analysis) for a multivalued relation, in which to a z there corresponds more than one w.

EXAMPLE 2 Real and Imaginary Parts of a Function

Find the real and imaginary parts of the functions: (**a**) $f(z) = z^2 - (2+i)z$ and (**b**) $g(z) = z + 2\,\mathrm{Re}(z)$.

Solution In each case, we replace the symbol z by $x + iy$, then simplify.

(**a**) $f(z) = (x+iy)^2 - (2+i)\,(x+iy) = x^2 - 2x + y - y^2 + (2xy - x - 2y)\,i$.
So, $u(x, y) = x^2 - 2x + y - y^2$ and $v(x, y) = 2xy - x - 2y$.

(**b**) Since $g(z) = x + iy + 2\,\mathrm{Re}(x+iy) = 3x + iy$, we have $u(x,\ y) = 3x$ and $v(x,\ y) = y$.

A function f can be defined without using the symbol z.

Every complex function is completely determined by the real functions $u(x, y)$ and $v(x, y)$ in (2). Thus, a complex function $w = f(z)$ can be defined by arbitrarily specifying two real functions $u(x, y)$ and $v(x, y)$, even though $w = u + iv$ may not be obtainable through familiar operations performed solely on the symbol z. For example, if we take, say, $u(x, y) = xy^2$ and $v(x, y) = x^2 - 4y^3$, then $f(z) = xy^2 + i(x^2 - 4y^3)$ defines a complex function. In order to find the value of f at the point $z = 3+2i$, we substitute $x = 3$ and $y = 2$ into the expression for f to obtain $f(3 + 2i) = 3 \cdot 2^2 + i(3^2 - 4 \cdot 2^3) = 12 - 23i$.

We note in passing that complex functions defined in terms of $u(x, y)$ and $v(x, y)$ can always be expressed, if desired, in terms of operations on the symbols z and $\bar{z}$.

The function e^z defined by:

$$e^z = e^x \cos y + ie^x \sin y \tag{3}$$

is called the **complex exponential function.**

EXAMPLE 3 Values of the Complex Exponential Function

Find the values of the complex exponential function e^z at the following points.

(a) $z = 0$ (b) $z = i$ (c) $z = 2 + \pi i$

Solution In each part we substitute $x = \text{Re}(z)$ and $y = \text{Im}(z)$ into (3) and then simplify.

(a) For $z = 0$, we have $x = 0$ and $y = 0$, and so $e^0 = e^0 \cos 0 + ie^0 \sin 0$. Since $e^0 = 1$ (for the real exponential function), $\cos 0 = 1$, and $\sin 0 = 0$, $e^0 = e^0 \cos 0 + i \sin 0$ simplifies to $e^0 = 1$.

(b) For $z = i$, we have $x = 0$ and $y = 1$, and so:

$$e^i = e^0 \cos 1 + ie^0 \sin 1 = \cos 1 + i \sin 1 \approx 0.5403 + 0.8415i.$$

(c) For $z = 2+\pi i$, we have $x = 2$ and $y = \pi$, and so $e^{2+\pi i} = e^2 \cos \pi + ie^2 \sin \pi$. Since $\cos \pi = -1$ and $\sin \pi = 0$, this simplifies to $e^{2+\pi i} = -e^2$.

$$z = re^{i\theta}. \tag{4}$$

We call (4) the **exponential form** of the complex number z.

if z_1 and z_2 are complex numbers, then

$$e^0 = 1, \tag{5}$$

$$e^{z_1} e^{z_2} = e^{z_1 + z_2}, \tag{6}$$

$$\frac{e^{z_1}}{e^{z_2}} = e^{z_1 - z_2}, \tag{7}$$

$$(e^{z_1})^n = e^{nz_1} \text{ for } n = 0, \pm 1, \pm 2, \ldots . \tag{8}$$

Polar Coordinates Up to this point, the real and imaginary parts of a complex function were determined using the Cartesian description $x + iy$ of the complex variable z. It is equally valid, and, oftentimes, more convenient to express the complex variable z using either the polar form $z = r(\cos\theta + i\sin\theta)$ or, equivalently, the exponential form $z = re^{i\theta}$.

$w = f(z)$, if we replace the symbol z with $r(\cos\theta + i\sin\theta)$, then we can write this function as:

$$f(z) = u(r, \theta) + iv(r, \theta). \tag{9}$$

We still call the real functions $u(r,\ \theta)$ and $v(r,\ \theta)$ in (9) the real and imaginary parts of f, respectively. For example, replacing z with $r(\cos\theta + i\sin\theta)$ in the function $f(z) = z^2$, yields, by de Moivre's formula,

$$f(z) = (r(\cos\theta + i\sin\theta))^2 = r^2\cos 2\theta + ir^2\sin 2\theta.$$

Thus, using the polar form of z we have shown that the real and imaginary parts of $f(z) = z^2$ are

$$u(r, \theta) = r^2\cos 2\theta \quad \text{and} \quad v(r, \theta) = r^2\sin 2\theta, \tag{10}$$

As with Cartesian coordinates, a complex function can be defined by specifying its real and imaginary parts in polar coordinates. The expression $f(z) = r^3 \cos\theta + (2r\sin\theta)i$, therefore, defines a complex function. To find the value of this function at, say, the point $z = 2i$, we first express $2i$ in polar form:

$$2i = 2\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right).$$

We then set $r = 2$ and $\theta = \pi/2$ in the expression for f to obtain:

$$f(2i) = (2)^3\cos\frac{\pi}{2} + \left(2\cdot 2\sin\frac{\pi}{2}\right)i = 8\cdot 0 + (4\cdot 1)i = 4i.$$

Complex Functions as Mappings

Recall that if f is a real-valued function of a real variable, then the graph of f is a curve in the Cartesian plane. Graphs are used extensively to investigate properties of *real* functions

we discuss the concept of a *complex mapping*, which was developed by the German mathematician Bernhard Riemann to give a geometric representation of a complex function. The basic idea is this. Every complex function describes a correspondence between points in two copies of the complex plane. Specifically, the point z in the z-plane is associated with the unique point $w = f(z)$ in the w-plane. We use the alternative term **complex mapping** in place of "complex function" when considering the function as this correspondence between points in the z-plane and points in the w-plane. The geometric representation of a complex mapping $w = f(z)$ attributed to Riemann consists of two figures: the first, a subset S of points in the z-plane, and the second, the set S' of the images of points in S under $w = f(z)$ in the w-plane.

S is a set of *complex* numbers. And a **function** *f* defined on *S* is a rule that assigns to every *z* in *S* a complex number *w*, called the *value* of *f* at *z*. We write

$$w = f(z).$$

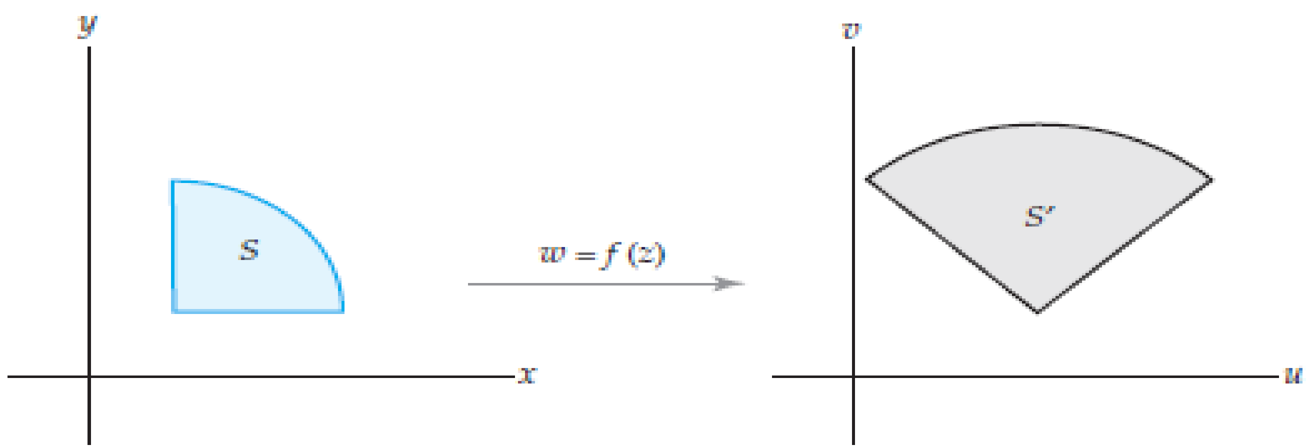
Here *z* varies in *S* and is called a **complex variable**. The set *S* is called the *domain of definition* of *f* or, briefly, the **domain** of *f*. (In most cases *S* will be open and connected, thus a domain as defined before.)

Example: $w = f(z) = z^2 + 3z$ is a complex function defined for all *z*; that is, its domain *S* is the whole complex plane.

The set of all values of a function *f* is called the **range** *of f*.

We assume all functions to be ***single valued relations***, as usual: to each z in S there corresponds but *one* value $w = f(z)$ (but, of course, several z may give the same value $w = f(z)$, just as in calculus). Accordingly, we shall *not use* the term "multivalued function" (used in some books on complex analysis) for a multivalued relation, in which to a z there corresponds more than one w.

If the set S has additional properties, such as S is a domain or a curve, then we also use symbols such as D and D' or C and C', respectively, to denote the set and its image under a complex mapping. The notation $f(C)$ is also sometimes used to denote the image of a curve C under $w = f(z)$.



(a) The set S in the z-plane

(b) The image of S in the w-plane

The image of a set S under a mapping $w = f(z)$

Mappings A useful tool for the study of real functions in elementary calculus is the graph of the function. Recall that if $y = f(x)$ is a real-valued function of a real variable x, then the graph of f is defined to be the set of all points $(x, f(x))$ in the two-dimensional Cartesian plane. An analogous definition can be made for a complex function. However, if $w = f(z)$ is a complex function, then both z and w lie in a complex plane. It follows that the set of all points $(z,\ f(z))$ lies in four-dimensional space (two dimensions from the input z and two dimensions from the output w). Of course, a subset of four-dimensional space cannot be easily illustrated. Therefore:

We cannot draw the graph a complex function.

The concept of a complex mapping provides an alternative way of giving a geometric representation of a complex function.

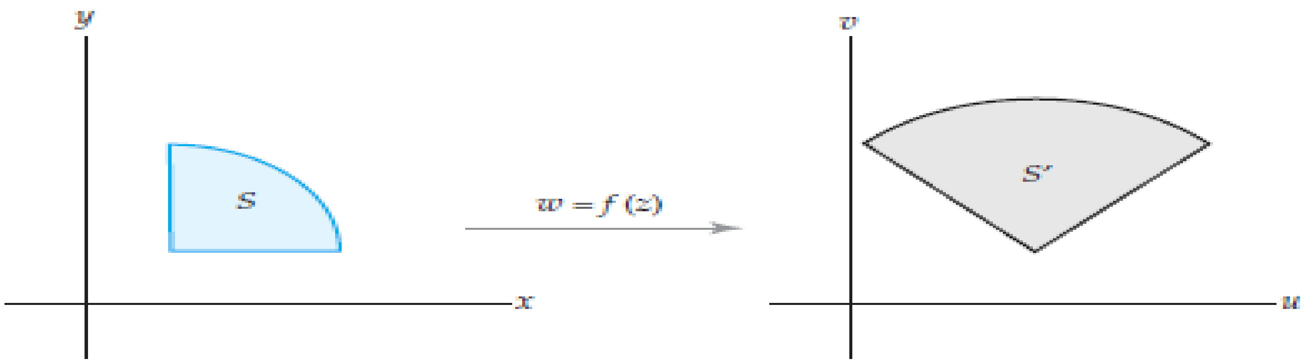
we use the term **complex mapping** to refer to the correspondence determined by a complex function $w = f(z)$ between points in a z-plane and images in a w-plane. If the point z_0 in the z-plane corresponds to the point w_0 in the w-plane, that is, if $w_0 = f(z_0)$, then we say that f **maps** z_0 onto w_0 or, equivalently, that z_0 is **mapped** onto w_0 by f.

As an example of this type of geometric thinking, consider the real function $f(x) = x + 2$. Rather than representing this function with a line of slope 1 and y-intercept $(0, 2)$, consider how one copy of the real line (the x-line) is mapped onto another copy of the real line (the y-line) by f. Each point on the x-line is mapped onto a point two units to the right on the y-line (0 is mapped onto 2, while 3 is mapped onto 5, and so on). Therefore, the real function $f(x) = x + 2$ can be thought of as a mapping that *translates* each point in the real line two units to the right.

On order to create a geometric representation of a complex mapping, we begin with two copies of the complex plane, the z-plane and the w-plane, drawn either side-by-side or one above the other. A complex mapping is represented by drawing a set S of points in the z-plane and the corresponding set of images of the points in S under f in the w-plane.

If $w = f(z)$ is a complex mapping and if S is a set of points in the z-plane, then we call the set of images of the points in S under f the ***image of S under f****, and we denote this set by the symbol S'.*

If the set S has additional properties, such as S is a domain or a curve, then we also use symbols such as D and D' or C and C', respectively, to denote the set and its image under a complex mapping. The notation $f(C)$ is also sometimes used to denote the image of a curve C under $w = f(z)$.



(a) The set S in the z-plane **(b)** The image of S in the w-plane

The image of a set S under a mapping $w = f(z)$

(1)

a) Find the image of $A(2,1)$ and $B(0,-1)$ under the complex mapping $w=z^2$.

b) If you go from A to B with line segment, how to go from A' to B'; here A' and B' are the images of A and B, respectively.

A' and B' are in w-plane

A and B are in z-plane

c) Find the image of $x=1$ under the complex mapping $w=z^2$

d) Find the image of $y=3x$ under the complex mapping $w=z^2$

a) $\omega = f(z) = u(x,y) + i v(x,y) = z^2 = (x+iy)^2$

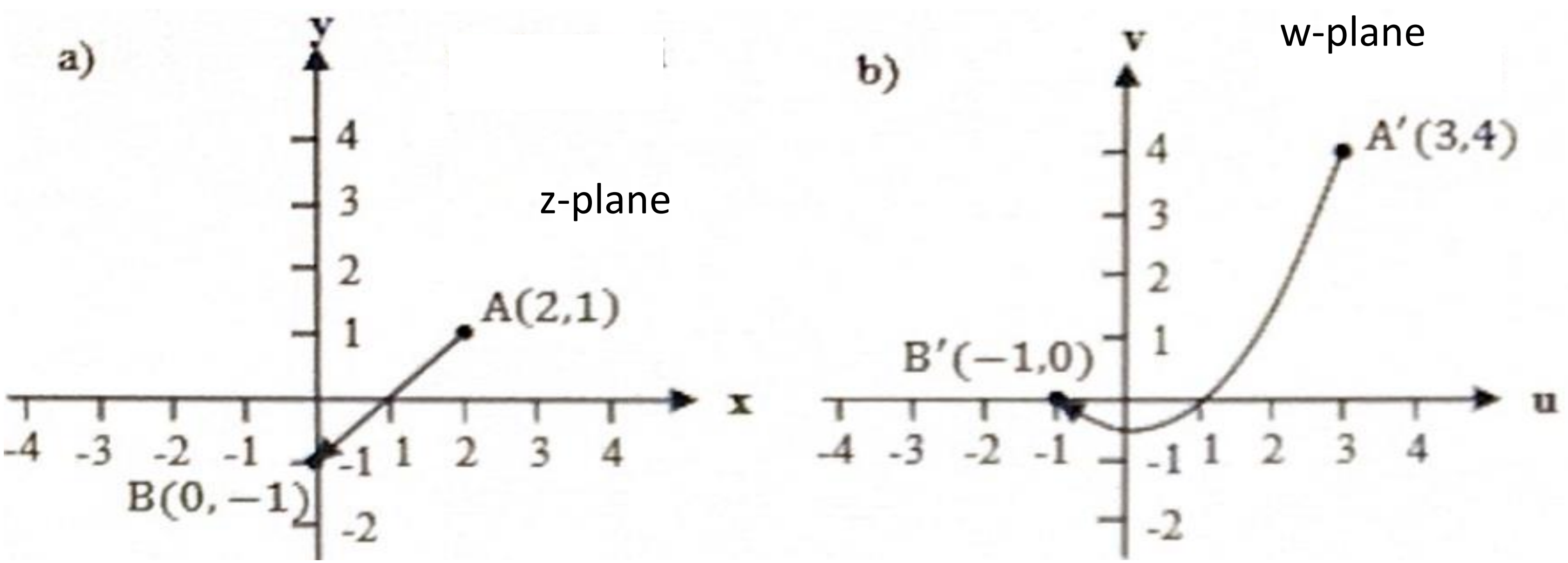
$$= x^2 - y^2 + i\,2xy$$

for the point A, $u(x,y) = x^2 - y^2 = 3$; $v(x,y) = 2xy = 4$

$$\boxed{A' = A'(3,4)}$$

for the point B, $u(x,y) = x^2 - y^2 = -1$; $v(x,y) = 2xy = 0$

$$\boxed{B' = B'(-1,0)}$$



b) The equation of AB is $y=x-1$. By using this equation

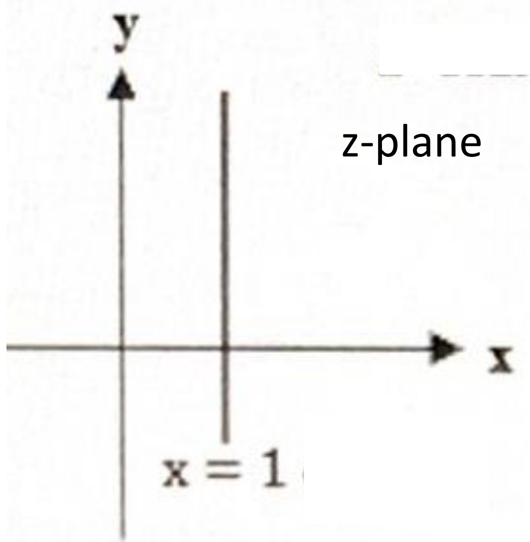
$$\left.\begin{aligned} u(x,y) &= x^2-y^2 = x^2-(x-1)^2 = 2x-1 \\ v(x,y) &= 2xy = 2x(x-1) \end{aligned}\right\} \text{solve it}$$

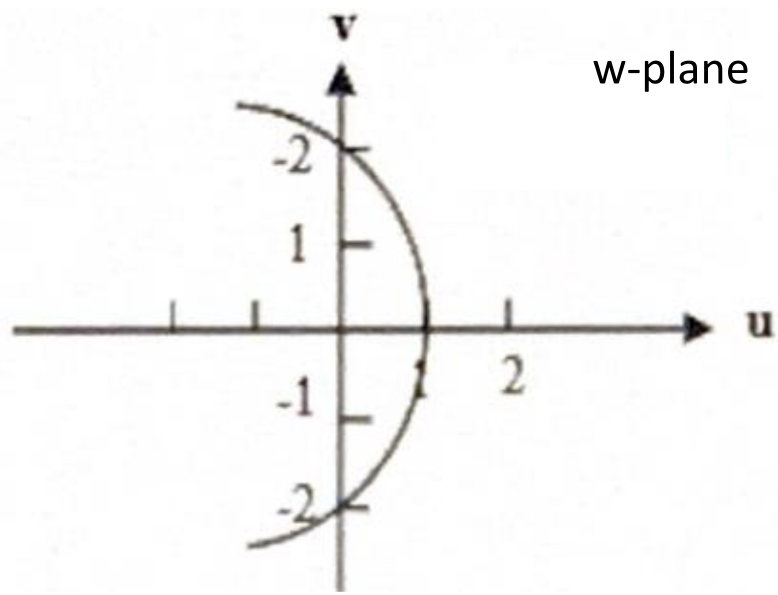
$V=\frac{u^2}{2}-\frac{1}{2}$ is found and it is a parabola passing through A' and B' in uv-plane.

c) $u(x,y)=x^2-y^2$, $v(x,y)=2xy \Rightarrow y=\frac{v}{2x}$

$$u=x^2-\frac{v^2}{4x^2}$$

and for $x=1$ line, $u=1-\frac{v^2}{4}$ parabola is obtained.

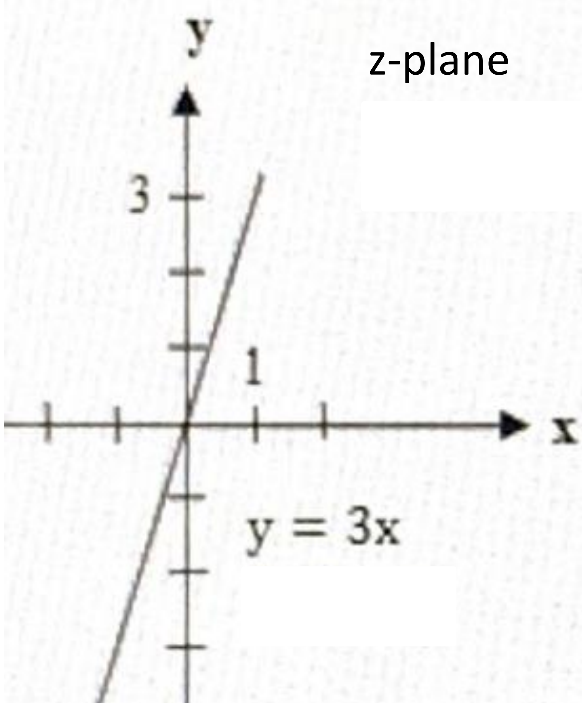


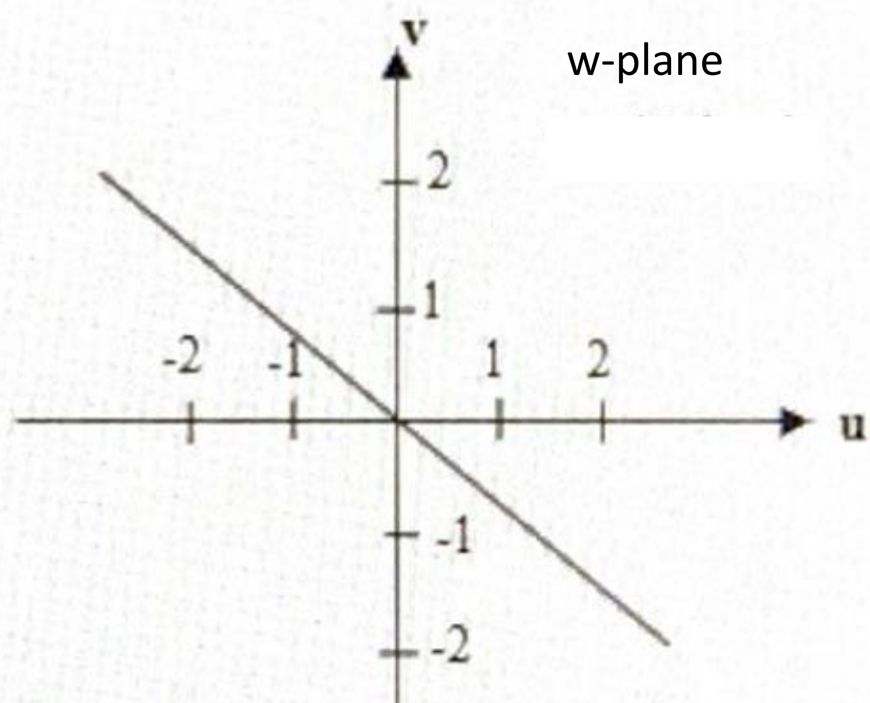


d) for $y=3x$, $u = x^2 - y^2 = x^2 - 9x^2 = -8x^2 \Rightarrow$

$$x^2 = -\frac{u}{8}$$

$$v = 2xy = 2x(3x) = 6x^2$$
$$= 6\left(-\frac{u}{8}\right)$$
$$= -\frac{3}{4}u$$





(2)

a) Find the image of $z=-1+i$ under the complex mapping $w=(1-i)z+1-2i$

b) Find the image of $\operatorname{Im}(z)>1$ under the complex mapping $w=(1-i)z+1-2i$

Solution:

a) $w=(1-i)(-1+i)+1-2i=-1+i+i+1+1-2i=1$

b) $w=(1-i)z+1-2i$ can be written as

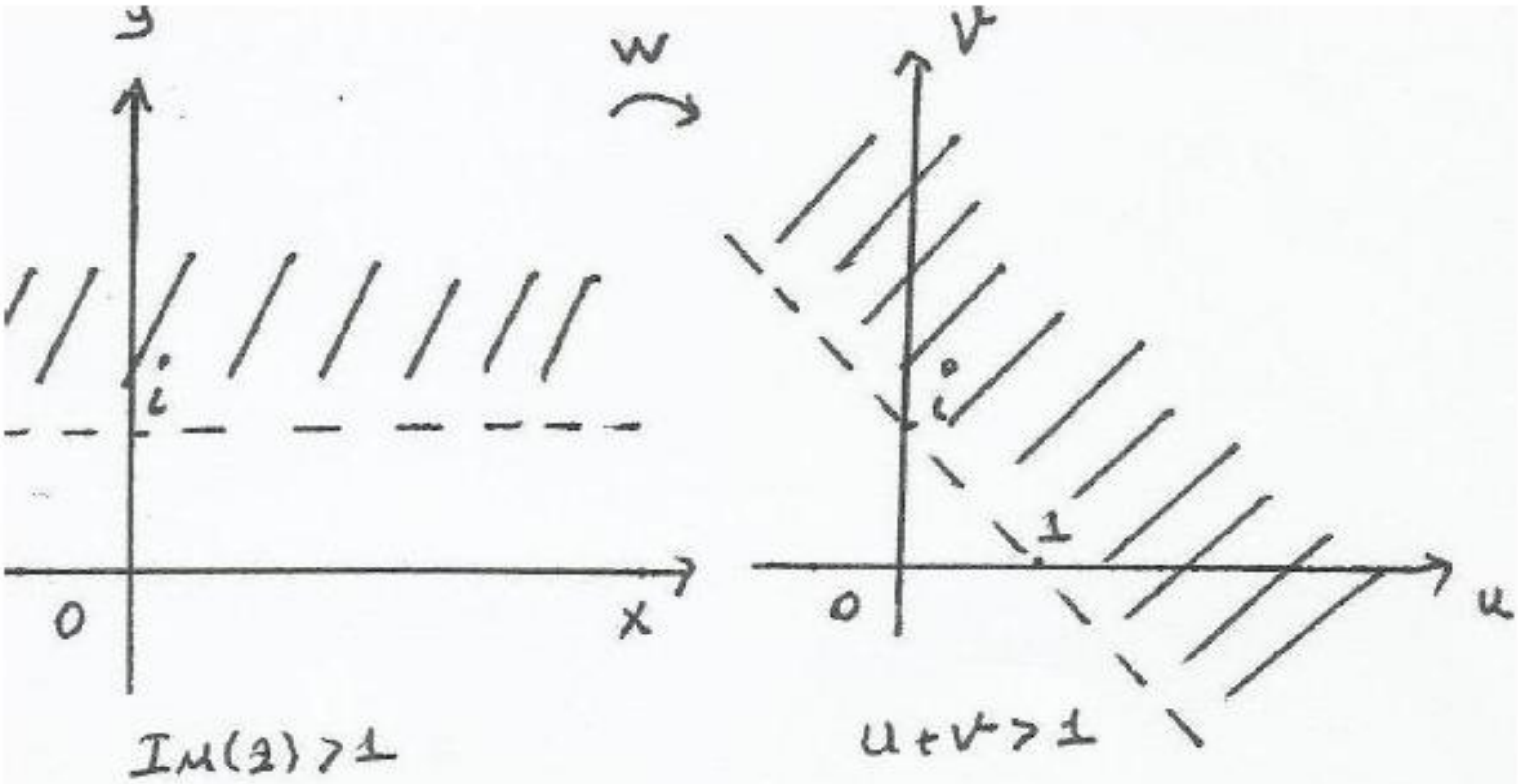
$z=\dfrac{w-1+2i}{1-i}$. Here, $z=x+iy$ $w=u+iv$

$$x+iy=\frac{u+iv-1+2i}{1-i}$$

$$\operatorname{Re}(z)=x=\frac{u-v-3}{2} \qquad \operatorname{Im}(z)=y=\frac{u+v+1}{2}$$

the image of $\operatorname{Im}(z)>1$, $\operatorname{Im}(z)=\dfrac{u+v+1}{2}>1$

$$\Rightarrow \quad u+v+1>2$$
$$u+v-1>0$$
$$u+v>1$$



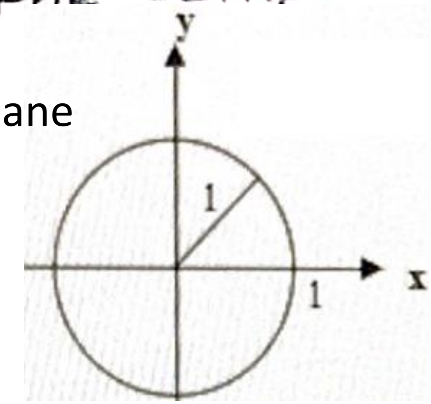
y
w
v
i
i
1
0
x
0
u
Im(z) > 1
u + v > 1

(3)

a) Find the image of $|z|=1$ under the complex mapping $w=\frac{1}{z}$

b) Find the image of $y=\frac{1}{2}$ under the complex mapping $w=\frac{1}{z}$

z-plane



Solution:

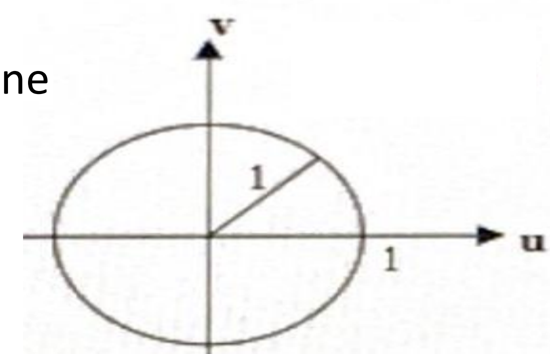
a) $|z|=(z.\bar{z})^{1/2}=\sqrt{x^2+y^2}=1$

$$w=\frac{1}{z}=\frac{1}{x+iy}=\frac{x-iy}{(x+iy)(x-iy)}=\frac{x-iy}{x^2+y^2}$$

$$u(x,y)=\frac{x}{\underbrace{x^2+y^2}_{1}}=x \qquad v(x,y)=\frac{-y}{\underbrace{x^2+y^2}_{1}}=-y$$

$$\boxed{u^2+v^2=x^2+y^2=1}$$

w-plane



b) $u = \dfrac{x}{x^2+y^2} \Rightarrow x^2+y^2 = \dfrac{x}{u}$ or $x = u(x^2+y^2)$ (i)

$$v = \frac{-y}{x^2+y^2} = -\frac{y}{\frac{x}{u}} = -\frac{\overset{\to \frac{1}{2}}{y}}{x}u = -\frac{1}{2x}u$$

$$\Rightarrow x = -\frac{u}{2v} \quad (ii)$$

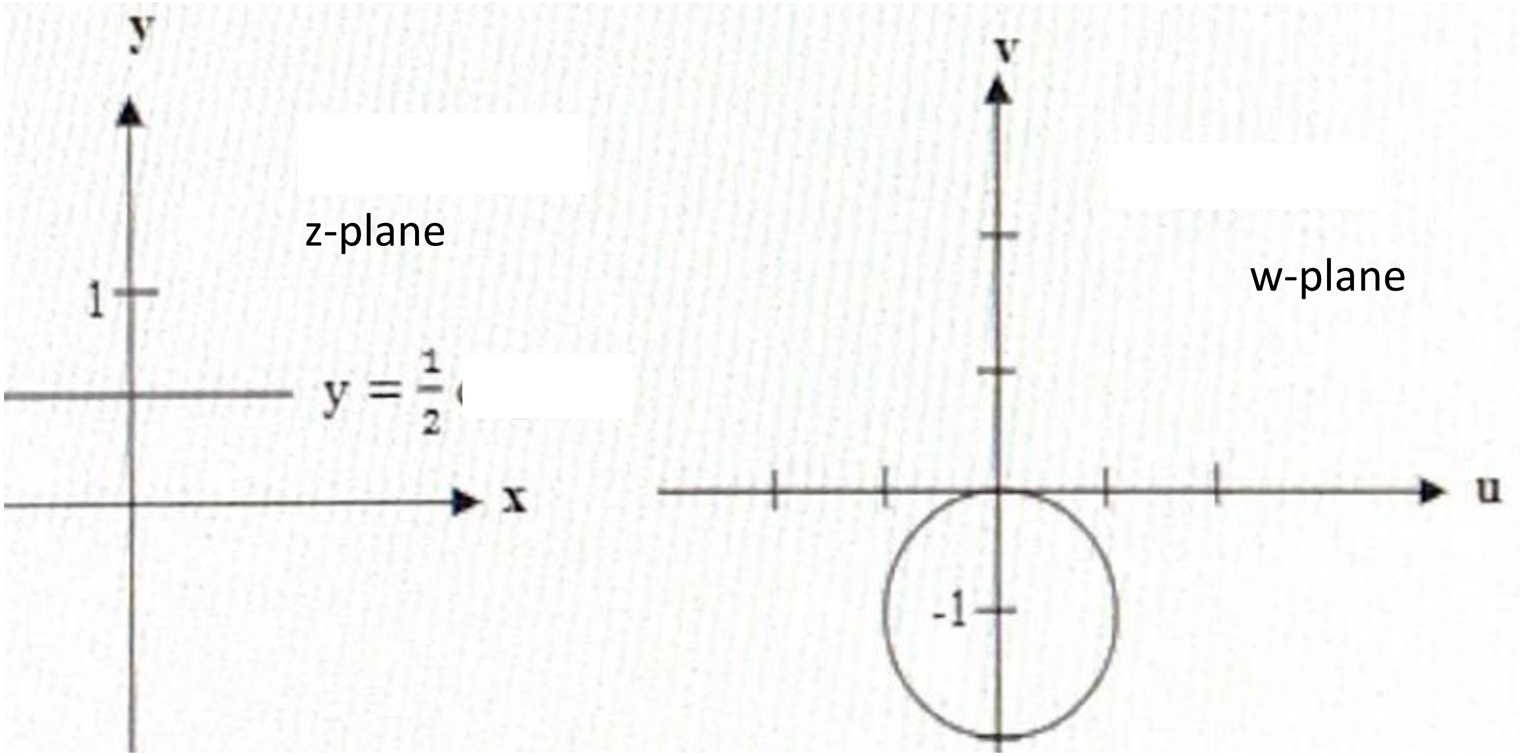
from (i) and (ii)

$$u(x^2+y^2) = -\frac{u}{2v}$$

$$\cancel{u}\left(\frac{u^2}{4v^2}+\frac{1}{4}\right) = -\frac{\cancel{u}}{2v} \Rightarrow \frac{u^2}{2v^2}+\frac{1}{2} = -\frac{1}{v}$$

$$u^2+v^2+2v=0$$

$$u^2+(v+1)^2=1$$



y
z-plane
1
$y = \frac{1}{2}$
x
v
w-plane
u
-1

(4)

For $w^4 = z$, find the image of $z = 1$.

$r = \sqrt{x^2 + y^2} = 1$, $\quad \theta = 0 + 2k\pi \quad (k = 0, 1, 2, 3)$

$w = z^{1/4} = (x + iy)^{1/4} = (re^{i\theta})^{1/4}$

$w = r^{1/4} e^{i\theta/4} = 1^{1/4} \cdot e^{i\left(\frac{\theta + 2k\pi}{4}\right)}$

$= \cos\left(\frac{2k\pi}{4}\right) + i \sin\left(\frac{2k\pi}{4}\right)$

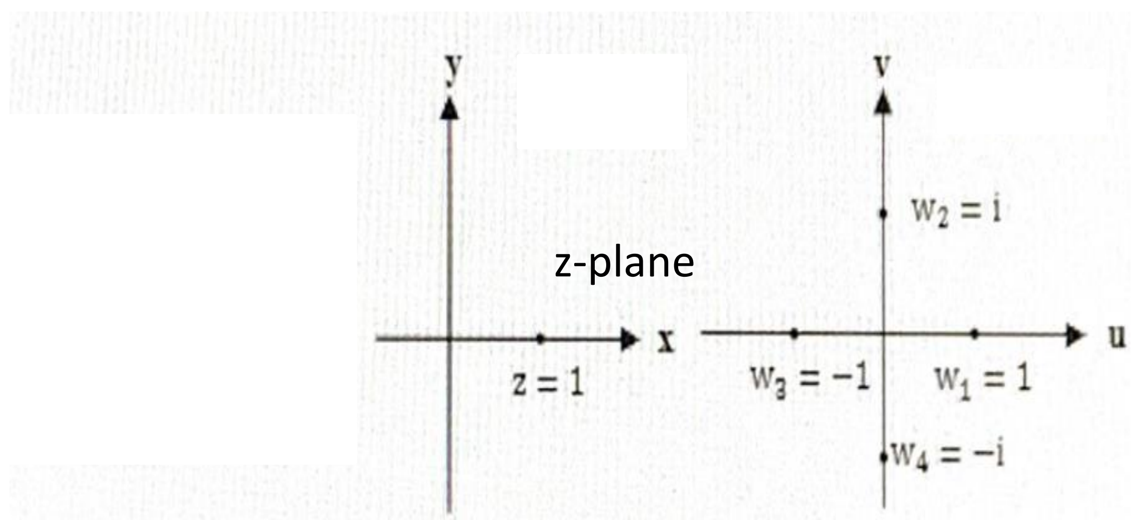
for $z = 1$; $\quad w = z^{1/4}$

for $k = 0$, $\omega_1 = 1$

for $k = 1$, $\omega_2 = i$

for $k = 2$, $\omega_3 = -1$

for $k = 3$, $\omega_4 = -i$



w-plane

EXAMPLE 1 **Image of a Half-Plane under $w = iz$**

Find the image of the half-plane $\text{Re}(z) \geq 2$ under the complex mapping $w = iz$ and represent the mapping graphically.

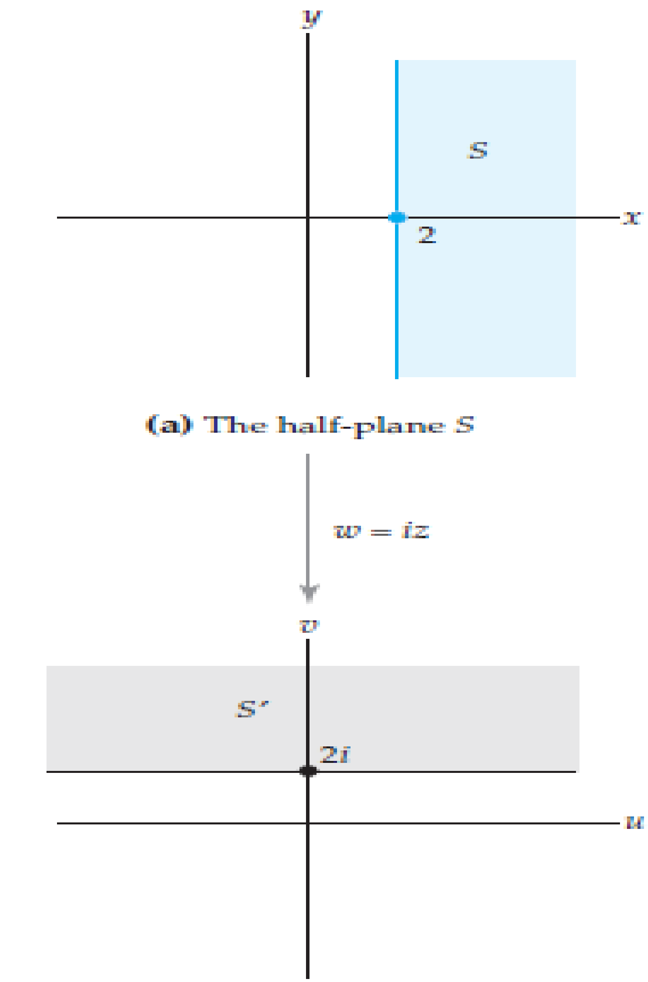
Solution Let S be the half-plane consisting of all complex points z with $\text{Re}(z) \geq 2$. We proceed as illustrated in Figure . Consider first the vertical boundary line $x = 2$ of S shown in color in Figure (a). For any point z on this line we have $z = 2 + iy$ where $-\infty < y < \infty$. The value of $f(z) = iz$ at a point on this line is $w = f(2 + iy) = i\,(2 + iy) = -y + 2i$. Because the set of points $w = -y + 2i$, $-\infty < y < \infty$, is the line $v = 2$ in the w-plane, we conclude that the vertical line $x = 2$ in the z-plane is mapped onto the horizontal line $v = 2$ in the w-plane by the mapping $w = iz$. Therefore, the vertical line shown in color in Figure (a) is mapped onto the horizontal line shown in black in Figure (b) by this mapping.

Now consider the entire half-plane S shown in color in Figure (a). This set can be described by the two simultaneous inequalities,

$$x \geq 2 \quad \text{and} \quad -\infty < y < \infty. \tag{1}$$

In order to describe the image of S, we express the mapping $w = iz$ in terms of its real and imaginary parts u and v; then we use the bounds given by (1) on x and y in the z-plane to determine bounds on u and v in the w-plane. By replacing the symbol z by $x + iy$ in $w = iz$, we obtain $w = i(x + iy) = -y + ix$, and so the real and imaginary parts of $w = iz$ are:

$$u(x,\ y) = -y \quad \text{and} \quad v(x,\ y) = x. \tag{2}$$



(a) The half-plane S

(b) The image, S', of the half-plane S

From (1) and (2) we conclude that $v \geq 2$ and $-\infty < u < \infty$. That is, the set S', the image of S under $w = iz$, consists of all points $w = u + iv$ in the w-plane that satisfy the simultaneous inequalities $v \geq 2$ and $-\infty < u < \infty$. In words, the set S' consists of all points in the half-plane lying on or above the horizontal line $v = 2$. This image can also be described by the single inequality $\mathrm{Im}(w) \geq 2$. In summary, the half-plane $\mathrm{Re}(z) \geq 2$ shown in color in Figure (a) is mapped onto the half-plane $\mathrm{Im}(w) \geq 2$ shown in gray in Figure (b) by the complex mapping $w = iz$.

The mapping $w = iz$

EXAMPLE 2 Image of a Line under $w = z^2$

Find the image of the vertical line $x = 1$ under the complex mapping $w = z^2$ and represent the mapping graphically.

Solution Let C be the set of points on the vertical line $x = 1$ or, equivalently, the set of points $z = 1 + iy$ with $-\infty < y < \infty$. We proceed as in Example 1. the real and imaginary parts of $w = z^2$ are $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$, respectively. For a point $z = 1 + iy$ in C, we have $u(1, y) = 1 - y^2$ and $v(1, y) = 2y$. This implies that the image of S is the set of points $w = u + iv$ satisfying the simultaneous equations:

$$u = 1 - y^2 \tag{3}$$

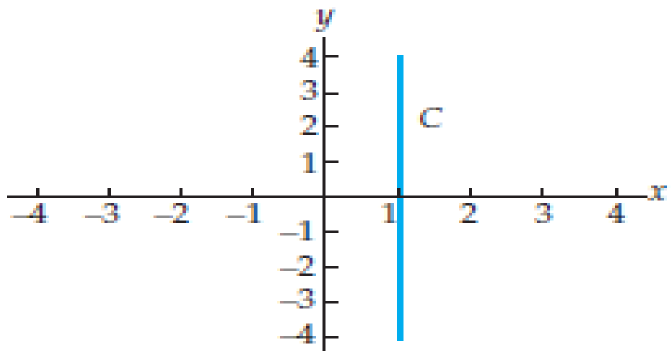
and

$$v = 2y \tag{4}$$

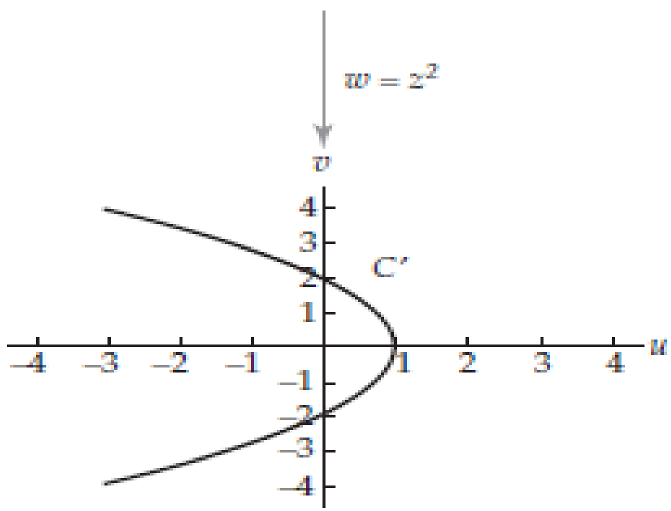
for $-\infty < y < \infty$. Equations (3) and (4) are *parametric equations* in the real parameter y, and they define a curve in the w-plane. We can find a Cartesian equation in u and v for this curve by eliminating the parameter y. In order to do so, we solve (4) for y and then substitute this expression into (3):

$$u = 1 - \left(\frac{v}{2}\right)^2 = 1 - \frac{v^2}{4}. \tag{5}$$

Since y can take on any real value and since $v = 2y$, it follows that v can take on any real value in (5). Consequently, C'—the image of C—is a parabola in the w-plane with vertex at (1,0) and u-intercepts at $(0, \pm 2)$. See Figure (b). In conclusion, we have shown that the vertical line $x = 1$ shown in color in Figure (a) is mapped onto the parabola $u = 1 - \frac{1}{4}v^2$ shown in black in Figure (b) by the complex mapping $w = z^2$.



(a) The vertical line $\mathrm{Re}(z) = 1$



(b) The image of C is the parabola $u = 1 - \frac{1}{4}v^2$

The mapping $w = z^2$

Parametric Curves in the Complex Plane For a simple complex function, the manner in which the complex plane is mapped might be evident after analyzing the image of a single set, but for most functions an understanding of the mapping is obtained only after looking at the images of a variety of sets. We can often gain a good understanding of a complex mapping by analyzing the images of *curves* (one-dimensional subsets of the complex plane) and this process is facilitated by the use of parametric equations.

If $x = x(t)$ and $y = y(t)$ are real-valued functions of a real variable t, then the set C of all points $(x(t),\ y(t))$, where $a \leq t \leq b$, is called a **parametric curve**. The equations $x = x(t)$, $y = y(t)$, and $a \leq t \leq b$ are called **parametric equations** of C. A parametric curve can be regarded as lying in the complex plane by letting x and y represent the real and imaginary parts of a point in the complex plane. In other words, if $x = x(t)$, $y = y(t)$,

and $a \leq t \leq b$ are parametric equations of a curve C in the Cartesian plane, then the set of points $z(t) = x(t) + iy(t),\ a \leq t \leq b$, is a description of the curve C in the complex plane. For example, consider the parametric equations $x = \cos t$, $y = \sin t$, $0 \leq t \leq 2\pi$, of a curve C in the xy-plane (the curve C is a circle centered at (0,0) with radius 1). The set of points $z(t) = \cos t + i \sin t$, $0 \leq t \leq 2\pi$, describes the curve C in the complex plane. If, say, $t = 0$, then the point $(\cos 0, \sin 0) = (1, 0)$ is on the curve C in the Cartesian plane, while the point $z(0) = \cos 0 + i \sin 0 = 1$ represents this point on C in the complex plane. This discussion is summarized in the following definition.

In a parametrization $z(t) = x(t) + iy(t)$, t is a real variable.

Parametric Curves in the Complex Plane

If $x(t)$ and $y(t)$ are real-valued functions of a real variable t, then the set C consisting of all points $z(t) = x(t) + iy(t)$, $a \leq t \leq b$, is called a **parametric curve** or a **complex parametric curve**. The complex-valued function of the real variable t, $z(t) = x(t) + iy(t)$, is called a **parametrization** of C.

Properties of curves in the Cartesian plane such as continuous, differentiable, smooth, simple, and closed can all be reformulated to be properties of curves in the complex plane. These properties are important in the study of the complex integral

Two of the most elementary curves in the plane are lines and circles. Parametrizations of these curves in the complex plane can be derived from parametrizations in the Cartesian plane. It is also relatively easy to find these parametrizations directly by using the geometry of the complex plane. For example, suppose that we wish to find a parametrization of the line in the complex plane containing the points z_0 and z_1.

$z_1 - z_0$ represents the vector originating at z_0 and terminating at z_1, shown in color in Figure . If z is any point on the line containing z_0 and z_1, then inspection of Figure indicates that the vector $z - z_0$ is a real multiple of the vector $z_1 - z_0$. Therefore, if z is on the line containing z_0 and z_1, then there is a real number t such that $z - z_0 = t\,(z_1 - z_0)$. Solving this equation for z gives a parametrization $z(t) = z_0 + t\,(z_1 - z_0) = z_0\,(1 - t) + z_1 t$, $-\infty < t < \infty$, for the line. Note that if we restrict the parameter t to the interval $[0, 1]$, then the points $z(t)$ range from z_0 to z_1, and this gives a parametrization of the line segment from z_0 to z_1. On the other hand, if we restrict t to the interval $[0, \infty]$, then we obtain a parametrization of the ray emanating from z_0 and containing z_1. These parametrizations are included in the following summary.

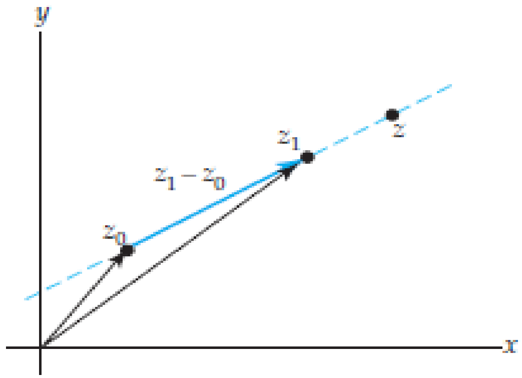


Figure Parametrization of a line

Common Parametric Curves in the Complex Plane

Line

A parametrization of the line containing the points z_0 and z_1 is:

$$z(t) = z_0(1 - t) + z_1 t, \quad -\infty < t < \infty. \tag{6}$$

Line Segment

A parametrization of the line segment from z_0 to z_1 is:

$$z(t) = z_0(1 - t) + z_1 t, \quad 0 \leq t \leq 1. \tag{7}$$

Ray

A parametrization of the ray emanating from z_0 and containing z_1 is:

$$z(t) = z_0(1 - t) + z_1 t, \quad 0 \leq t < \infty. \tag{8}$$

Circle

A parametrization of the circle centered at z_0 with radius r is:

$$z(t) = z_0 + r\left(\cos t + i \sin t\right), \quad 0 \leq t \leq 2\pi. \tag{9}$$

In exponential notation, this parametrization is:

$$z(t) = z_0 + re^{it}, 0 \leq t \leq 2\pi. \tag{10}$$

Image of a Parametric Curve under a Complex Mapping

If $w = f(z)$ is a complex mapping and if C is a curve parametrized by $z(t)$, $a \leq t \leq b$, then

$$w(t) = f(z(t)), \quad a \leq t \leq b \tag{11}$$

is a parametrization of the image, C' of C under $w = f(z)$.

EXAMPLE 3 Image of a Parametric Curve

Use (11) to find the image of the line segment from 1 to i under the complex mapping $w = \overline{iz}$.

Solution Let C denote the line segment from 1 to i and let C' denote its image under $f(z) = \overline{iz}$. By identifying $z_0 = 1$ and $z_1 = i$ in (7), we obtain a parametrization $z(t) = 1 - t + it$, $0 \leq t \leq 1$, of C. The image C' is then given by (11):

$$w(t) = f(z(t)) = \overline{i(1 - t + it)} = -i(1 - t) - t, \quad 0 \leq t \leq 1.$$

With the identifications $z_0 = -i$ and $z_1 = -1$ in (7), we see that $w(t)$ is a parametrization of the line segment from $-i$ to -1. Therefore, C' is the line segment from $-i$ to -1.

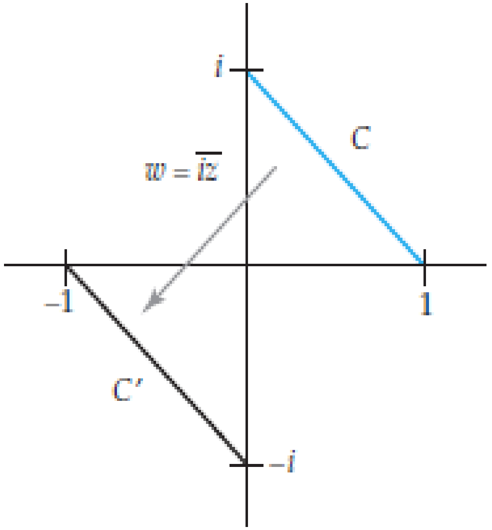


Figure : The mapping $w = \overline{iz}$

EXAMPLE 4 Image of a Parametric Curve

Find the image of the semicircle shown in color in Figure under the complex mapping $w = z^2$.

Solution Let C denote the semicircle shown in Figure and let C' denote its image under $f(z) = z^2$. We proceed as in Example 3. By setting $z_0 = 0$ and $r = 2$ in (10) we obtain the following parametrization of C:

$$z(t) = 2e^{it}, \quad 0 \leq t \leq \pi.$$

Thus, from (11) we have that:

$$w(t) = f(z(t)) = \left(2e^{it}\right)^2 = 4e^{2it}, \quad 0 \leq t \leq \pi, \tag{12}$$

is a parametrization of C'. If we set $t = \frac{1}{2}s$ in (12), then we obtain a new parametrization of C':

$$W(s) = 4e^{is}, \; 0 \leq s \leq 2\pi. \tag{13}$$

From (10) with $z_0 = 0$ and $r = 4$, we find that (13) defines a circle centered at 0 with radius 4. Therefore, the image C' is the circle $|w| = 4$. We represent this mapping in Figure using a single copy of the plane. In this figure, the semicircle shown in color is mapped onto the circle shown in black by $w = z^2$.

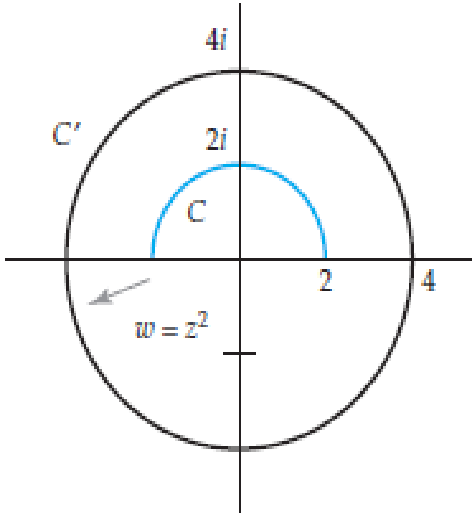


Figure The mapping $w = z^2$

Complex mappings are closely related to parametric curves in the plane. In later sections, we use this relationship to help visualize the notions of limit, continuity, and differentiability of complex functions. Parametric curves will also be of central importance in the study of complex integrals much as they were in the study of real line integrals.